DC Space <-> eRoomReserve

Integration Contract

Version 1.0 ? August 23, 2026
DC Space (MongoDB) ? eRoomReserve (Firebase)

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1. Purpose

Connect DC Space and eRoomReserve without sharing databases. DC Space sends on-campus event room requests; eRoomReserve checks availability and performs first-level room approval; DC Space admin gives final event sign-off.

2. System Ownership

DC Space (MongoDB: dcspace_user, dcspace_admin) owns users, events, registrations, attendance, and certificates.

eRoomReserve (Firebase) owns rooms, availability, and the reservation approval workflow.

Neither system reads or writes the other's primary database directly.

3. Integration Flow

1. Organizer submits On Campus event in DC Space (status: pending).
2. DC Space writes reservation to eRoom Firebase (status: pending).
3. Organizer opens eRoomReserve (optional deep link).
4. eRoomReserve checks availability and approves or rejects the room.
5. eRoomReserve calls the DC Space webhook with the status update.
6. DC Space admin reviews the event and gives final approve/reject.
7. Event goes live for students.

4.1 DC Space -> eRoomReserve (Firebase Write)

DC Space upserts linkedUsers and reservations using a Firebase service account provided by eRoomReserve.

Collection: linkedUsers (document ID = normalized email)

Required fields: dcSpaceUserId, email, fullName, sourceSystem (always "dcspace"), updatedAt.

Optional: firstName, lastName, studentNumber, role, organizationPart, organizationRole, school, course.

Collection: reservations (document ID = dcspace-{eventId})

DC Space writes: reservationId, dcSpaceEventId, eventTitle, startAt, endAt, requestedByEmail, requestedByName, venueType (On Campus), conceptPaperUrl, conceptPaperName, status (pending).

eRoomReserve writes on approval: roomId, roomName, building, approvedBy, approvedAt, status (approved).

eRoomReserve writes on rejection: rejectionReason, status (rejected).

Collection: rooms (eRoom-owned)

Fields: roomId, roomName, building, capacity, isActive.

4.2 eRoomReserve -> DC Space (HTTPS Webhook)

Endpoint: POST {DC_SPACE_BASE_URL}/api/integrations/iroom/webhook

Header: X-Iroom-Secret: {shared secret}

Header: Content-Type: application/json

Valid status values: pending, approved, rejected, cancelled

Approved payload example:


```
  {
    "reservationId": "dcspace-674abc123",
    "dcSpaceEventId": "674abc123",
    "status": "approved",
    "roomId": "avr-101",
    "roomName": "AVR Room 101",
    "building": "Main Building",
    "approvedBy": "admin@sdca.edu.ph",
    "approvedAt": "2026-08-23T08:00:00.000Z"
}
```


Rejected payload example:


```
  {
    "reservationId": "dcspace-674abc123",
    "dcSpaceEventId": "674abc123",
    "status": "rejected",
    "rejectionReason": "No rooms available for the requested time slot."
}
```


Responses: 200 success ? 401 unauthorized ? 400 validation error

eRoomReserve must call this webhook when a room is approved, rejected, or cancelled.

4.3 DC Space Endpoints (Reference)

```
POST /api/integrations/iroom/reservations - DC Space creates or refreshes a reservation.
GET /api/integrations/iroom/reservations?eventId= - DC Space reads room status.
POST /api/integrations/iroom/webhook - eRoomReserve pushes status updates.
```

5. Deep Link (Organizer Handoff)

When configured, DC Space opens:

```
{IROOM_APP_URL}?reservationId=dcspace-{eventId}&dcSpaceEventId={eventId}&source=dcspace
```

Concept paper URL:

```
{DC_SPACE_BASE_URL}/api/events/{eventId}/attachments/concept-paper
```

6. Credentials to Exchange

From eRoomReserve to DC Space: Firebase Project ID, Service Account JSON, IROOM_APP_URL, shared webhook secret.

From DC Space to eRoomReserve: production webhook URL, staging webhook URL (optional), DC Space base

URL.

DC Space .env keys:

```
IROOM_FIREBASE_PROJECT_ID
IROOM_FIREBASE_SERVICE_ACCOUNT
IROOM_APP_URL
IROOM_WEBHOOK_SECRET
IROOM_USERS_COLLECTION=linkedUsers
IROOM_RESERVATIONS_COLLECTION=reservations
```

7. Approval Responsibilities

Stage 1 - Room request: DC Space submits event and writes reservation to Firebase.
Stage 2 - Availability + room approval: eRoomReserve checks slots, assigns room, sends webhook.
Stage 3 - Final event approval: DC Space admin approves or rejects for publication.

For on-campus events, DC Space admin final approval should only happen after eRoomReserve sets status: approved.

8. Security Requirements

All production traffic over HTTPS.
Webhook authenticated via X-Iroom-Secret header.
Firebase service account scoped to linkedUsers and reservations writes only.
Do not sync passwords, auth tokens, attendance, registrations, or full media files.
Recommend webhook retries with exponential backoff (max 5 attempts).

9. Status Lifecycle

none -> pending -> approved (happy path)

```
               -> rejected (no room / policy)
               -> cancelled (event cancelled in either system)
```

eRoom Firebase: reservations.status
DC Space Mongo: events.iroomStatus, events.iroomRoomId, events.iroomRoomName

10. eRoomReserve Delivery Checklist

[ ] Firebase linkedUsers + reservations collections created
[ ] Service account issued to DC Space team
[ ] Reservation review UI (filter sourceSystem = dcspace)
[ ] Room availability check for startAt / endAt
[ ] Approve/reject workflow with room assignment
[ ] Webhook caller to DC Space on every status change
[ ] Deep-link page for reservationId + dcSpaceEventId
[ ] Handle cancelled when DC Space cancels/postpones event
[ ] Staging environment tested end-to-end

11. Contacts

DC Space backend: ____________________________

eRoomReserve backend: _______________________

Shared secret owner: ________________________

12. References

docs/IROOM_INTEGRATION.md

app/api/integrations/iroom/webhook/route.ts

lib/iroom/sync.ts

Signatures

DC Space: Name / Date _______________________

eRoomReserve: Name / Date ___________________